

An Agile Method, A Contractual Relationship, and Distance

AN AGILE METHOD, A CONTRACTUAL RELATIONSHIP, AND DISTANCE: A TRIAD OF CHALLENGING CONDITIONS FOR A SUCCESSFUL SYSTEM DEVELOPMENT PROJECT

Carmen Bernier, Ph.D.
Laval University
Carmen.Bernier@fsa.ulaval.ca

Line Dubé, Ph.D. and Vital Roy, Ph.D.
HEC Montréal

ABSTRACT

This study explores how a thoughtful tailoring of an agile method contributed to the successful development of a complex and unique information system in a context where teams work across organizational and geographical boundaries. More specifically, the findings show how a large consulting firm managed to overcome the ever-elusive business requirement and distance challenges in a contractual relationship with a global client. The tailoring efforts addressed the challenges of working under a fixed-price contractual agreement by managing a total effort umbrella and by continuously aligning the functionalities to be delivered in the overall project with the work to be realized in each sprint; the challenges of working at a distance were met by adding time to the typical one-month sprint, and by developing a rich portfolio of communication tools and practices; and finally, the challenges of working with a global client were mainly dealt with by adopting a “just enough” strategy where a smart communication-oriented documentation was introduced. The findings show that the tailoring efforts addressed the agility needs of the project and resulted in a successful endeavor for all parties¹.

KEY WORDS:

Information System Development; Agile Scrum Method; Method Tailoring; Agility.

INTRODUCTION

Information system developments (ISD) take place in a very competitive and rapidly changing business environment. Consequently, development methods are under pressure to “react to market dynamics, respond and incorporate changing customer requirements, and also deal with everyday technological innovations” (Sarker and al., 2009, p. 286). Complaints about traditional plan-based software development approaches are legion: system development takes too long, costs too much, and the end result is often of low quality in terms of performance and its capacity to fulfill users’ changing needs (Fitzgerald et al., 2006; Moore et al., 2007; Berczuk, 2007; Lindvall et al., 2002). Austin and Devin (2009) note that “[c]ritics of plan-based approaches argue that to be sufficiently flexible, especially in increasingly turbulent environments, a software process needs to revisit specifications and refine requirements even late in development” (p. 463).

To deal with the challenges of increasingly turbulent business environments, new “agile” development methods, such as Scrum and eXtreme Programming (XP), have been proposed. These new methods are said to be particularly well suited for contexts where requirements change during the development life cycle (Rising and Janoff, 2000; Fitzgerald et al., 2006, Petersen and Wohlin, 2009). Agile methods are based on a set of values and principles promoted through the “Manifesto for Agile Software Development” published in 2001. According to Vlaanderen et al. (2011), the introduction of agile principles in ISD represented a major innovation. However, Vidgen and Wang (2009) report that “despite the popularity of agile methods in software development, there is debate about what agility is and how it is achieved. The debate suffers from a lack of understanding of agile concepts and how agile software development is practiced” (p. 355). Despite this confusion, Baskerville et al. (2011) affirm that “even though agile ISD might still (from a research and theory-building point of view) be conceptually weak, our findings show that in practice the concept is now distinct and well defined enough for practitioners to explicitly explain how they use and combine the agile approach” (p. 544). In a similar line of thoughts, many authors in the research community have made a plea for more rigorous empirical evidence to understand agile methods in general (e.g., Conboy and Fitzgerald, 2007), but also to understand how they can be adapted to address the specific needs of different project contexts, such as globally distributed software development (e.g., Holmström et al., 2006; Ramesh et al., 2006) and mission-critical software development (e.g., Boehm, 2002; Cockburn, 2002; Turk et al., 2005), aiming at a better understanding of their benefits, challenges, domains of applicability and tailoring (Lindvall et al., 2002; MacCormack et al., 2003; Erickson et al., 2005; Mann and Maurer, 2005; Abrahamsson et al., 2003; Turk et al., 2005).

In light of the above, and as recommended by Vidgen and Wang (2009), our objective in the current study is to contribute to the goal of building a solid cumulative knowledge base leading to an increased understanding of how agile software development is used in practice. More specifically, we explore the following two questions: 1) how can an agile method be tailored to successfully support the development of an information system when challenging conditions — namely a contractual consultant-client relationship and working at a distance with a global client — are present?, and 2) is a tailored agile method still agile? These questions are increasingly important and relevant in the evolving context of IS development involving different partners, clients and consulting firms, becoming global and geographically dispersed (Holström et al., 2006). Very few evidence is available on how an agile method can be successfully tailored to support system development in such a context.

To investigate these research questions, we adopted a case study approach which enabled a rich understanding of how the agile methodology Scrum was tailored in the context of the Knowledge+ project. From a research standpoint, this project is a particularly interesting case since its basic characteristics, namely a contractual relationship, distance, and a global client, are a far departure from the ideal setting promoted by Scrum protagonists. Drawing upon empirical experiences is a key contribution that informs the IS domain (Berger, 2007).

The article is organized as follows. The next section investigates the literature on the main characteristics of agile methods, and more specifically Scrum, and presents a summary of the

challenges most commonly encountered. The literature review is followed by a presentation of the research method. The next section describes the Knowledge+ project and the organization involved followed by a presentation and a discussion of the findings. Finally, a conclusion presents the implications for research and practice, and through exposing the limitations of the study, proposes avenues for future research.

THEORETICAL BACKGROUND

The agile Scrum method and its applicability challenges

Scrum was first proposed in 1995 and is presented as a flexible, adaptable, empirical, productive and iterative method (Schwaber and Beedle, 2002). According to Vlaanderen et al. (2011), “central to Scrum is the idea that that many of the processes during development cannot be predicted. The only two parts that are fully defined during a software development project are the first and last phase (planning and closure). In between, the final product is developed by several teams in a series of flexible black boxes called ‘sprints’” (p. 59). Each sprint iteration includes three phases. The first phase, a planning phase (sprint planning), is used to establish the requirements (backlog) representing the functionalities that the team will be developing during the iteration, to evaluate the modifications required to the system architecture, and to develop the high-level design. The client, called the product owner, is free to choose any set of functionalities, as long as the development team agrees that they can be developed during a normal iteration. The client usually creates stories from which use cases will be compiled. The second phase is a coding and test phase (sprint) for the development of the negotiated new functionalities, with a constant and clear focus on providing an application that responds to the client’s needs. Contrary to traditional plan-driven methods, where the client control progress based on “paper”, these short cycles and the focus on working code allow the client to “see” what has been done during a cycle. Finally, during the closure phase, the development team presents the new functionalities to the product owner in an event called the “sprint review”, accompanied by the final documentation, the results of the testing process, and the release itself.

Scrum is very popular and is often considered the most “agile” (Qumer and Henderson-Sellers, 2008). Performing especially well when requirements are unknown, unknowable, or changing (Schwaber and Beedle, 2002), Scrum can lead to significant advantages due to better knowledge transfer and early frequent feedback (Petersen and Wohlin, 2009). For agile tenants, “[a]gile’ is not just a matter of process structure but is also a set of closely associated values, principles, and practices” (Austin and Devin, 2009, p. 475). The agile manifesto (Beck et al., 2001) clearly exposes the underlying precepts: 1) Individuals and interactions over processes and tools; 2) Working software over comprehensive documentation; 3) Client collaboration over contract negotiation; and 4) Responding to change over following a plan. These precepts raise some concerns, notably concerning the applicability of the methods to a wide variety of software development scenarios and to diversified environments. Table 1 presents an overview of the agile precepts, their related practices, and their applicability challenges.

Agile Manifesto Precepts	Related Practices	Applicability Challenges
1. Individuals and interactions over processes and tools	Team members work in an open area for sustained overall collaboration and coordination is fluid and supported by the shared knowledge and face-to-face interactions (Moore et al., 2007; Turk et al., 2005). Close, face-to-face interactions between the development team and the client (Boehm, 2002; Cockburn, 2002; Turk et al., 2005).	• Distributed software development (Ramesh et al., 2006; Carmel, 1999; Turk et al., 2005). • The client may have other important matters to attend to, or may be miles away (Turk et al., 2005; William et al., 2007).
2. Working software over comprehensive documentation	The output of the software development process is a piece of working code, not documentation (Theunissen et al., 2003; Cockburn, 2002; Qumer and Henderson-Sellers, 2008; Turk et al., 2005).	• Preserving critical knowledge through time (Neill, 2003; De Souza et al., 2005). • Managing the legal relationship with consultants (Turk et al., 2005; Fruhling and De Vreede, 2006). • Fulfilling the quality system requirements such as ISO or CMM (Boehm, 2002; McMichael and Lombardi, 2007).
3. Client collaboration over contract negotiation	Contracts loosely, informally, and continuously negotiated (Ramesh et al., 2006).	• Contract negotiation when the output is not clearly defined (Cockburn, 2002). • Fostering a productive vendor/client relationship (Eckfeldt et al., 2005). • Scope and risk management (Turk et al., 2005; Eckfeldt et al., 2005).
4. Responding to change over following a plan	Short iterative cycles called sprint where responsive development has precedence over detailed long-term planning (Vlaanderen et al., 2011).	• Controlling progress from the client's point of view (Schwaber, 2004).

Table 1: Overview of the Agile Precepts, their Related Practices, and their Applicability Challenges

The “individuals and interactions over processes and tools” precept emphasizes the necessity for team members to work in an open area in anticipation of an increase in energy level, an enhanced focus and cohesiveness, and a more sustained overall collaboration (Moore et al., 2007). Coordination is fluid and supported by the shared knowledge of what the others are doing. Because of the increased potential for miscommunication and delays, face-to-face interactions have precedence over mediated communication (Turk et al., 2005). In Scrum, for example, a mandatory face-to-face 15-minute daily meeting is organized each morning. This precept seems to take collocation for granted; however, it is increasingly difficult to achieve as distributed software development becomes more common (Ramesh et al., 2006; Carmel, 1999; Turk et al., 2005).

Agile methods also advocate the same close, face-to-face interactions between the development team and the client. The client's representative should be “committed, knowledgeable,

collaborative, representative and empowered” (Boehm, 2002, p.66). Continuous feedback from the client is seen as crucial to success. The client’s representative should be close by, ready and able to give input and to make decisions that may be needed in order for the team to progress (Cockburn, 2002; Turk et al., 2005). This on-going interaction provides the development team with a deeper understanding of the users’ needs and habits (Cockburn, 2002).

Though this may be the ideal, in practice, it may not always be possible. The client may have other important matters to attend to (Turk et al., 2005), or may be miles away (Williams et al., 2007). According to Ramesh et al. (2006), important challenges in agile distributed development are related to communication needs. Reasonable arrangements need to be made to ensure that developers have constant access to the client’s feedback. Some authors (e.g., Fruhling and De Vreede, 2006; Cockburn, 2002) have presented some possible alternatives, such as weekly meetings, weekly interview sessions with users, ethnographic studies of the user community, surveys, friendly alpha-test groups, etc., to keep an open communication flow between the development team and the client.

The precept pertaining to “working software over documentation” drives many debates in the literature about the importance of documentation in software development and maintenance (Theunissen et al., 2003; Lindvall et al., 2002; Fruhling and De Vreede, 2006; Lethbridge et al., 2003; Neill, 2003). Most agile method advocates insist that the output of the software development process is a piece of working code, not documentation. The basic tenet is that no or little time should be spent on documentation that does not provide direct value to the project’s main goal (Theunissen et al., 2003). Agilists mainly raise two arguments to support their point: (1) nothing can replace face-to-face communication for the transmission of tacit knowledge (Cockburn, 2002; Qumer and Henderson-Sellers, 2008); and (2) the most reliable source of information about what a system does is the code itself, since documentation is typically never updated (Turk et al., 2005). The absence of formal documentation poses the problem of how to preserve critical knowledge through time (De Souza et al., 2005), especially when people leave (Neill, 2003), or the application is developed by a consultant. Different forms of documentation may be essential in a distributed environment. A legal relationship may also make the exchange of some formal documentation mandatory (Turk et al., 2005; Fruhling and De Vreede, 2006).

The absence of documentation also raises the issue of agile method compatibility with ISO or CMM processes (Boehm, 2002; McMichael and Lombardi, 2007). But by definition, agile methods are highly adaptable and should be able to comply with required standards (Theunissen et al., 2003), including those pertaining to the production of documentation. Organizations seem to be able to integrate their agile efforts into the requirements of their quality systems (McMichael and Lombardi, 2007; Vriens, 2003; Fitzgerald et al., 2006). In fact, most authors would agree that the “no documentation” principle can hardly be applied. The challenge is rather to identify what is “just enough” documentation, what knowledge should be codified and what may remain tacit (Heeager, 2012; Nerur et al., 2005; Cockburn, 2002).

In an agile environment, the emphasis is on “client collaboration over contract negotiation”. In such a context, contracts are supposed to be loosely, informally, and continuously negotiated

(Ramesh et al., 2006). This does not fit well with the long-established transaction cost perspective where a contract is awarded on a predefined set of commitments and for a fixed price. Consulting firms formally define contractual relationships with clients in order to manage the scope of the project and the required resources in order to ensure a reasonable profit margin. Such contracts usually carry a clause stipulating that any changes to the contract will entail significant cost increases (Turk et al., 2005). The scope-related risks are typically shouldered by the vendor who tries to attenuate them by factoring in a high contingency. Overall, this practice makes the project extremely expensive (Eckfeldt et al., 2005).

The precept of “responding to change over following a plan” therefore puts pressure on how a contract is negotiated. While several authors discuss system development by consultants using agile methods (e.g., Cockburn, 2002), very few examine the contractual issues of such relationships and how to make them work. As mentioned by Eckfeldt et al. (2005), it is difficult to find clear guidelines to fashion a contractual agreement that will foster a productive vendor/client relationship. As an alternative, these authors propose a target-cost contract, which, given a flexible scope, includes a labor rate, and fixed contingency and profit margins. This encourages both parties to find the most effective and efficient solution, and to drop or reduce unnecessary functionalities. This type of contract has the advantage of eliminating the win-lose situation of the traditional fixed-cost contract and to make the vendor and the client work together to develop the best solution in the shortest time frame (Eckfeldt et al., 2005).

Method Tailoring

The ideal setting for the use of an agile method such as Scrum — a small, self-organizing, collocated team of fewer than 20 developers closely working with one on-site client over 30-day cycles — is seldom encountered. In Scrum, for example, each individual practice (collocated development team, rapid client feedback, a streamlined decision process, and the focus on developing software) has to be present if a “working piece of software” is to be delivered at the end of each month. However, some authors have shown that several benefits can be realized even if some of the practices have to be modified, loosened or ignored (Fruhling and De Vreede, 2006; Berczuk, 2007). In practice, Petersen and Wohlin (2009) note that “... from our experience in industry we know that methods presented in books are often tailored to specific needs and that practices are enriched or left out as they do not fit into the context” (p. 1488). While method tailoring is common in software development projects and can be done successfully (Fitzgerald et al., 2006), selecting the right combination of individual agile practices that will preserve the benefits of agility is not easy (Qumer and Henderson-Sellers, 2008). The question of preserving the benefits of agility can be posed as expressed by Lewis and Neher (2007): “When is a duck NOT a duck?” The anecdotal answer may lie in the words of Cockburn (2002, p.195): “Consider *agile* as an attitude, not a formula.”

Defining “agility” as an attitude is of utmost interest, but not an easy task. In this study, we build on Sarker and Sarker’s (2009) definition of agility as a multifaceted concept to explore the relationships between the Scrum tailoring efforts in the Knowledge+ project and their impacts on the project agility. Sarker and Sarker (2009, p.440) maintain that in a distributed software development context, the concept of agility includes three different dimensions. *Resource agility*

“is based on the distributed development team’s access to necessary human and technological resources”. *Process agility* “pertains to the agility that originates in the team’s systems development method guiding the project, its environmental scanning, and sense-making routines to anticipate possible crises, and its work practices enabling collaboration across time zones”. Finally, *linkage agility* “arises from the nature of interactional relationships within the distributed team and with relevant project stakeholders, and is composed of cultural and communicative elements”.

RESEARCH APPROACH

Like many authors interested in agile development research (e.g., Vidgen and Wang, 2009; Berger, 2007; McAvoy and Butler, 2009; Doherty et al., 2012; Heeager, 2012), and in order to develop a rich understanding of how an agile method can be successfully tailored to meet challenging contextual conditions, we used a case study approach, which provides rich insights into the project’s human, social, and organizational contexts.

The case under study, the Knowledge+ project, was an ISD project executed by a Canadian branch of a large global IT consulting firm (ranked by Forbes Global among the top 20 in the computer services industry) for a large American client, with offices around the world. Following Mannz et al.’s (2011) logic, the study of this case allows an investigation of the practices of an exemplar global IT consulting firm. The Knowledge+ project was chosen because it was unanimously considered a success and it represented a first attempt by this firm to innovatively test the applicability of an agile approach to a software development project meeting challenging conditions, namely a contractual relationship, distance, elusive requirements, and a global client. The researchers were granted full access, which resulted in a rich telling of the project and its context.

The client considered the Knowledge+ system to be developed a strategic asset. It was unique and had to be tailored to its specific needs for the company to compete more effectively in a complex and highly competitive financial environment. The consulting firm was reluctant to tackle the project on a fixed-price and fixed-delivery-date basis (the standard in this industry) since the client had just been through a two-year period where the requirements had revealed themselves to be impossible to pin down. The consulting team proposed a different approach, namely the use of a tailored agile approach (largely inspired by Scrum), which the client enthusiastically accepted as a possible answer to its previous nightmares. No roadmap existed and the consultant’s team members had to design their “tailored” approach based on their combined expertise in software development and project management, and their knowledge of the newest ISD methodological approaches. The final proposition also had to take into consideration the client’s and the project’s structural, historical, and budget constraints. The expected profit margin did not allow for the consulting firm to move its team to the client’s site for the project duration, so the work had to be mainly realized with a consultant’s team and a client’s team geographically dispersed across two countries and three time zones.

Since we were interested in how the consultant had tailored its approach to the specific needs of the project, we chose to analyze the development project from the consultants’ point of view. The

client's team had little technology or project management expertise (and specifically no expertise in agile methodologies) and the consultant single-handedly led the methodology tailoring efforts. Moreover, the client evaluated the project as being highly satisfactory as documented following the consulting firm's ISO standards, which include the evaluation of both the development process and the system delivered. The best evidence however lies in the client's desire to add many sprint cycles to the original contractual agreement.

All things considered, this single revelatory case study provided a rich and invaluable opportunity to examine our research questions (Fahy et al., 2009; Benbasat et al., 1987; Yin, 2003) and to deepen our understanding of method tailoring under unconventional and challenging conditions.

Data collection

In this study, several sources were used to gather data. During a discussion with the project director and the account manager, the consultant's key players of the project were identified. The interviewees were selected so as to cover the key roles and actors in the project. Semi-structured face-to-face interviews were conducted with the account manager (1 interview), the project director (2 interviews), the Scrum Master who manages the Scrum process and who also serves as the senior functional architect on the project (2 interviews), and the technical architect (1 interview). This methodological approach is consistent with those adopted by Vidgen and Wang (2009) and Moe et al. (2010).

To get a thorough understanding of the Knowledge+ project, the interview guide was divided into three broad categories. The project description category included questions about the project's objectives, owner, organization, resources, risks and success factors. The second category, project development process and approach, included questions about the choice of the Scrum method, the challenges in its application to this specific project context, the changes that were made, and their impacts. The third category, project organization and management, included questions about how different the project was compared to the typical ISD project, the resources' roles and responsibilities, the key challenges in managing the project, and the project's and client's management and governance structure.

Each tape-recorded interview lasted from 60 to 90 minutes and was subsequently transcribed. Multiple e-mails and phone calls were also exchanged with each of the respondents to clarify specific elements. Confidential project documentation (the project proposal, the contract, the project planning, the output of the Scrum reviews, etc.) was also obtained. The software development practices in use in the project (events, actors, and practices) were extracted and the timeline plotted while looking for convergence among both the respondents, and the respondents and the documents. This on-going triangulation between two data sources and multiple respondents was used to lessen the presence of certain biases, such as social desirability or selective memory of events, consequently increasing the reliability and the general quality of the research.

Data analysis

Using the interview transcripts and the project documentation as a base, the project's software development practices were extracted and organized in a matrix and then compared to the prescribed Scrum practices as described by Schwaber (2004) and Schwaber and Beedle (2002). Reasons for the adaptations were then identified and investigated. Other issues pertaining to the applicability of agile methods, as raised in the literature review, were systematically investigated: quality assurance, contractual issues, and practices and technology specifically designed to work in a distributed fashion. The researchers' understanding of how the method was used and adapted was finally validated during two additional follow-up meetings with the account manager and the Scrum Master.

The next section presents the Knowledge+ project and describes the tailoring efforts that were implemented.

CASE DESCRIPTION

The Knowledge+ Project

Financials¹, the global client, is a world leader in the management of fund managers with employees in all major financial centers throughout the world. Located on the west coast of the United States, it had the very aggressive objective to triple its managed assets for its two core sectors: investment capital management and financial consulting services. Radical changes in some of the firm's operating methods were undertaken and the financial advisory service had to become increasingly productive. A new strategic information system called Knowledge+ was required to support Financials' agents and analysts in collecting, organizing, searching, and analyzing highly unstructured financial information from multiple varied sources (e.g., official prospectuses, historical data on value fluctuations, fund manager interviews and performance records). Once compiled by the new system, the data had to be reorganized into a user-friendly and reliable interface. This project was the largest and most complex IT development effort that Financials had ever undertaken with a price tag of about US\$4 million and a project duration of 12 months. The new system had to be developed using Microsoft .Net, a Web development framework Financials was not familiar with.

After two years of extensive efforts aimed at defining the business requirements, Financials had realized that they did not have neither the expertise nor the necessary resources to develop this complex system and hired ConsultCo to do so. Financials already had a successful ongoing working relationship with ConsultCo (*tier-1* outsourcing contract, including infrastructure management, server and data center management, and technical support center) and ConsultCo was widely known for its .Net and large project management expertise.

¹ All names are fictive to protect the anonymity of the organizations and people involved.

The Contractual Agreement

Considering the failure of Financials' previous attempt at producing a stable set of requirements for the Knowledge+ system, ConsultCo was reluctant to tackle the project on a fixed-price and fixed-delivery-date basis and to shoulder all the risks related to its elusive scope. ConsultCo therefore recommended using the Scrum agile development approach which Financials readily accepted. While ConsultCo would be responsible for the development of the system, the IT resources at Financials would be accountable for defining the business requirements and for conducting the user acceptance tests. A list of high level functionalities (e.g., reporting by product, data analysis by product) was established and used by ConsultCo to estimate the global effort required. In addition, the iterative approach based on Scrum that would be used was described in great detail (iterations, dates, roles and responsibilities, documents to be produced and approved, deliverables, meeting formats, etc.). The system was estimated to be a 3,000 person-day effort.

The Financials' team involved 17 user's representatives from different business units located around the world, including a super user who was IT knowledgeable and who acted as the official product owner and final arbiter for all business requirement issues and priorities. To this initial group, three IT business analysts and one database administrator were added. The contract stipulated that the representatives were to be available 50-75% of the time for face-to-face working sessions with the ConsultCo senior architects at the beginning of each sprint. The business analysts and other IT resources had to dedicate at least 75% of their time to the project. The ConsultCo's team included a senior functional architect, a senior technical architect, two functional analysts, and seven software developers working full time on the project. On specific occasions, several IT specialists joined the original team. The senior functional architect also played the role of Scrum Master.

The Development Approach – a Tailored Scrum

The Knowledge+ system was developed over a 12-month period. The first two months were used to set up the technical environment. The system was then developed during six overlapping two-month sprints (see Figure 1). At Financials' request, the last three months were dedicated to conducting additional integration tests and to providing support for the user training sessions.

The first month of each sprint was used to define the business requirements, thereby detailing the sought-after functionalities. During the first week of the cycle, ConsultCo's senior functional and technical architects met face-to-face with the client's team (user's representatives, business analysts, and DBA) at the client's site for three days in order to understand and validate the "existing process" document prepared by the client's business analysts in the previous month. Between scheduled meetings, ConsultCo's architects would prepare high level use cases (UML scripts format) for the required functionalities and evaluate the overall development effort. During the last of these face-to-face working sessions, the client's team and ConsultCo's architects finalized the list of functionalities to be developed during the next sprint. The product owner would then sign-off on this list. Upon their return home and over the following two weeks, ConsultCo's architects and analysts completed and documented the approved list of functionalities. They extended the use case first versions and developed the user interfaces while

working closely with the client's business analysts, by phone and video conference, to resolve any unexpected issues/questions. During the fourth week, via a collaborative environment tool (SharePoint), the complete functional documentation (use cases, user interfaces, class diagrams, data model, and business rules) was provided to the client's team for review and formal approval. At the end of the fourth week, typically after some minor adjustments, the functional documentation was approved by the product owner and became the official sprint backlog.

The second 30-day period of the sprint was used to code and test the approved functionalities. The software developers organized their work, validated their use case understanding with the architects, started the coding process and prepared the test scripts. All on-going work (updated functional documentation, code, daily build, issues and bug management, etc.) was recorded using collaborative working tools, to which all team members, consultant's and client's alike, had access. At the beginning of each day, ConsultCo's team members met face-to-face for a daily Scrum; no client's representatives attended this meeting. The project director prepared a standard weekly progress report that was sent to the project management offices of both the client's and the consulting firm's.

At the end of each sprint, the development team presented the production-ready sprint functionalities to Financials' team through a sprint review meeting organized through a Web application, along with an audio conference. The final use cases, system and integration test processes, updated bug list, data model and business rules were also included in the sprint delivery package.

In the end, ConsultCo's and Financials' teams were able to deliver a fully operational, tested and documented application to Financials' users. The project was such a success that the client immediately contracted ConsultCo for extra sprints with the objective to add enhanced functionalities to the Knowledge+ system.

FINDINGS AND DISCUSSION

An Agile Method for the Knowledge+ Project

On one hand, the Knowledge+ project seemed to have the basic characteristics that made it an appropriate candidate for an agile method. According to Applegate et al. (2007), it could be best described, from the client's point of view, as a large, low structure, low technological knowledge project, all characteristics pointing towards a high-risk project. After two years of intensive efforts, and the active intervention of outside specialists, Financials had not managed to produce a complete list of requirements, let alone the design specifications for the future system. The client had also elected to use a new technology with which it was unfamiliar. Because of strategic constraints, the project was also planned along a very tight 12-month schedule. Clearly the project was not well suited for a classic plan-driven approach: innovation and audacity were called for.

On the other hand, the Knowledge+ project was not going to be executed in the ideal setting for an agile method. The basic Scrum practices (a collocated development team, an on-site client, a continuous negotiation process, a streamlined decision process, and the focus on developing software) could not be easily implemented in the specific context of this project. This situation proved itself to be an opportunity for ConsultCo, who had trained personnel in the Scrum method, to market its technological and project management expertise. It was also a way to share some of the scope risks associated with the system's elusive nature with their client. But the Knowledge+ Project also raised concerns about how Scrum could be tailored to a software development context that was not in line with its central tenets and assumptions. Working under the umbrella of a fixed-price contractual agreement at a distance with a global client required major adaptations to the method's basic practices as presented in Table 2.

Working under a Fixed-Price Contractual Agreement Umbrella

A major challenge with an agile method such as Scrum is its limited support for subcontracting (Turk et al., 2005). Under normal circumstances, subcontracted tasks have to be well-defined, measureable and agreed upon by both parties. The subcontractor assumes the responsibility for delivering a fixed level of functionalities in a predefined period of time and for a set price, while the client has to define clear and complete specifications that the developers can understand and act upon.

Clearly, this was not the case in the Knowledge+ Project where both parties had to sign a contractual agreement. The elected strategy to overcome this difficulty was multifaceted. While the high-level list of functionalities may serve as a baseline for contract negotiation and thus make a fixed-price contract possible, it may also reveal itself to be a curse during the project. In Scrum, the product owner has full control over the product backlog. This backlog can evolve as the product owner sees fit. The development team decides with the product owner, in accordance with the development team's workload capacity, which elements of the product backlog will be included in the next sprint. In the fixed-price contract negotiated by Financials and ConsultCo, the list of functionalities to be delivered by the end of the project was fixed. By gauging the level of complexity of each functionality, ConsultCo was able to make a time estimate. However, throughout the project, each sprint initiation ended up being a mini-contract negotiation about the functionalities to be included, the level of sophistication awarded to each and how much effort

each one would require. Such negotiations called for compromises and give-and-take from both sides. While a “real” Scrum team just needs to be preoccupied by the feasibility of the sprint backlog, ConsultCo’s team members also had to keep their eyes on the overall list of functionalities identified in the initial contract. But the level of sophistication of each functionality had to be defined and agreed upon at sprint time and kept under control. Overall, despite a formal contractual agreement, the governance philosophy had to move from an arm’s-length contractual approach to one based on partnership and mutual trust. It was impossible for Financials and ConsultCo to negotiate a fully specified contract covering all aspects and imponderables of the future system. The only practical solution was to enter into a partnership arrangement. Partnerships can be described as a transactional/relational continuum (Paulin et al., 1997; Dwyer et al., 1987; Macneil, 1980). From the transaction costs perspective, contractual governance is concerned with understanding the specifics of setting up and implementing viable cooperation relationships, and delineating the terms and conditions (distribution of rights and duties between the partners) needed to organize such relationships. Relational governance is concerned with how independent firms align their shared processes on a day-to-day basis through organizational mechanisms with the aim of implementing the bilateral exchange of information (Goshal and Haspeslagh, 1990). The participants indeed acknowledged the fact that they could not spell out, in advance, the complete set of terms and conditions for the length of their forthcoming relationship. Rather, they opted for a tailored approach that allowed for periodic renegotiations of evolving requirements. They jointly defined how they would work in the contract.

Agile Precepts Derogation	Knowledge+ Project	Key Tailoring Efforts	Agility Dimensions (Sarker and Sarker, 2009)
<i>Working under a fixed-price contractual agreement umbrella</i>	Partnership arrangement and relational governance.	Financials and ConsultCo jointly defined how they would work in the contract. 1. By gauging the level of complexity, ConsultCo made a time estimate for each functionality to be delivered at the end of the project. Each sprint initiation ended up being a mini-contract negotiation about the functionalities to be included, the level of sophistication awarded to each and how much effort each one would require. The number of sprint and the number of hours per sprint were defined in the contract.	Linkage-based
<i>Working at a distance ...</i>	Project teams dispersed on two countries and three time zones.	2. The overlapping sprint cycles were extended to two months (one month to finalize the business requirements; one month for realization, testing, and delivery). 3. Teams agreed on a fixed period of the day where each would be available for telephone and e-mail communications. 4. Every month, ConsultCo sent the team's two senior architects to meet with the client's team on the west coast to discuss the content of the next sprint backlog. 5. Technology was used as a substitute for face-to-face sprint reviews. 6. The client insisted on having a set of fully documented functionalities before the development phase was launched (second month). 7. ConsultCo made good use of a permanent representative based at the client's office on the west coast. 8. For legal considerations and as per its ISO requirements, ConsultCo was eager to have documented and signed-off specifications that could be used as proof that the required functionalities had successfully been delivered.	Resource-based Process-based Linkage-based
<i>... with a global client</i>	The role of product owner for the global client was played by a group of user's representatives from Financials' various business units.	9. On the client's side, a super user had final authority on the group's decisions. This group met face-to-face for: - A three day-period (sometimes up to five) to define and agree on the list of functionalities for the following sprint. The consultant's senior architects were also physically present. - A week at the end of the design phase to approve the final functional packages (the communication with the consultant's architects were carried through electronic media). - Between those meetings, the business analysts were the official interfaces with the consultant's team.	Resource-based

Table 2: Overview of Findings – Tailoring Scrum for a Contractual Relationship and a Distributed Environment

Actions were initiated to support and nurture an atmosphere of trust between the partners. First and foremost, both companies relied on their accumulated shared experience (collaborating first on a *tier-1* outsourcing contract). In addition, in the year before the launch of the project, two

senior analysts from ConsultCo (now involved as architects in the current project) had worked full time with the client's team to help implement the .Net infrastructure. These analysts therefore had an intimate knowledge of Financials' needs and work environment and personally knew most of its IT team members.

Another important element that fostered Financials' trust was the presence of a sustained and open communication flow between the two teams. The setting up by Financials of a stable and representative user group that had full authority to decide on all functional issues pertaining to the project clearly contributed to the promptness and depth of communication between the two organizations. The timely delivery of anticipated bug-free functionalities at the end of each iteration, especially at the end of the first few sprints, also helped confirm Financials' choice of ConsultCo as a good faith and dependable partner that could be counted on to take up and support its objectives.

ConsultCo appreciated Financials' open-mindedness and maturity in accepting to let ConsultCo's architects evaluate the amount of work that could reasonably be realized during each cycle. During some iterations, ConsultCo underestimated the required effort and had to absorb the additional work as overtime; on other occasions, the development effort was less than planned and the developers had slack time that could be dedicated to other tasks. Both partners took for granted that, over the total duration of the project, things would average out. All in all, the willingness of both parties to embark in a real partnership agreement is probably the single most important factor that contributed to the successful completion of this project.

Working at a Distance ...

In the Knowledge+ project, the project team was dispersed on two distant locations: the client team on the west coast and the development team on the east coast, in different countries, more than 3,000 miles and three time zones away. This would be at odds with the Scrum method suggestion that the client be easily accessible for face-to-face meetings with the development team and for the review phase at the end of each cycle. Intense face-to-face communication is one of the method's basic building blocks, permitting mutual adjustments, tacit knowledge transmission and rapid reaction time. As noted by Turk et al. (2005), geographical distribution renders communications more difficult because of varying work schedules and differences in time zones, and because the participants cannot see each other and take advantage of the non-verbal dimension of their interactions. To overcome this handicap, ConsultCo had to bend the method's rules somewhat and implement operational and technological adaptations:

- The overlapping sprint cycles were extended to two months so as to allow the functional requirements to be included in the next sprint backlog to be documented (first month) and to give Financials the necessary time to examine the functional specifications and get everyone's comments before the final approval by the formal product owner. This served a dual purpose. On the one hand, the client insisted on having a set of fully documented functionalities before the development phase of a sprint was launched (at the beginning of the second month). The documentation was used as a discussion baseline for the client and the development teams. This was seen as a confidence building measure that granted a higher level of control to Financials on the project orientation. This written functional packages also

gave Financials a high level of assurance as to what could be expected at the end of the sprint. At the same time, for legal and ISO considerations, ConsultCo was eager to have documented and signed-off specifications that could be used as proof that the required functionalities had successfully been delivered. Ultimately, the written documentation proved to be an effective communication and coordination tool for both partners, reducing the need for face-to-face dialogue.

- In order to deal with the three-hour differential time zone lag, both teams agreed on a fixed period of the day where each would be available for telephone and e-mail communications.
- Because of their previous *tier-1* outsourcing relationship, ConsultCo had a permanent representative (client relationship manager) based at the client's office on the west coast, serving as the "eyes and ears" of the development team and providing continuous feedback.
- ConsultCo provided a Web video conferencing and sharing tool (WebEx, from Cisco), supporting an easy and efficient sharing of data, program code and application demonstrations. This technology was used as a substitute for face-to-face sprint reviews. SharePoint (Microsoft) was used for Word documentation (use cases), and Excel files for sprint backlog (updated daily by developers), project backlog, and diagram sharing.
- Each month, ConsultCo sent the two senior architects on the team to meet with the client's team on the west coast and discuss the content of the next sprint backlog. This activity was the closest approximation of the sprint meeting where the development team meets with the product owner and together they go through the product backlog items to identify which ones would be included in the next sprint, while taking into account the total time allowance.

In this working arrangement, all issues and questions involving the client (for example, a question about how to interpret a requirement) were answered promptly because of the enforced working rules and the technology used to communicate with the client's business analysts. Transparency between partners was established by providing Financials with a detailed weekly status report, that included progress tracked against plan, the burn-down chart, the current issues and risk log, and the following week's planned activities.

...With a Global Client

Working for a large and global client also carried its own set of challenges. The first one pertains to the identification and legitimacy of the product owner. In the Knowledge+ project context, the role of product owner was actually played by a group of specialists from Financials' various business units. The members of this group would meet at the beginning of each sprint. Their main responsibilities involved drafting the initial requirements, prioritizing, and approving the list of use cases resulting from the negotiation with ConsultCo's architects. This group was led by a super user, a key functional specialist with a solid expertise in financial information systems, who had final authority on the group's decisions. Since this group was not easily reachable throughout the development phase of a sprint cycle, the main contact points were Financials' IT business analysts who would answer the development team's questions in a timely fashion. The identity and roles of the product owner was therefore distributed and somewhat redefined.

As described, the sheer number of stakeholders involved on the client's side created a burden on the project and decreased the whole process pace. Typically, it took three days (sometimes up to five) to meet with the group of user representatives to finally get a list of approved functionalities

for the sprint to come. It also took a week at the end of the business requirement phase to get a final stamp of approval on the functional packages. While this process is important for the client's buy-in and for the formal approval called for by the legal and ISO requirements, it also heavily contributed to the lengthening of the sprint cycle and made the approach less receptive to up-to-the-moment changes. While in a standard 30-day sprint, a change request can easily be integrated into the following increment, the same change request in the Knowledge+ project occurring, for example, during the first sprint in the middle of month 3 (when the list of functionalities has already been established) would only be dealt with in the planning stage of sprint 2. The change requirement would therefore be developed and implemented at the end of the fifth month, which is over 75 days later. As commented by the Scrum Master, this aspect of the method "required some serious on-going education" of the client throughout the project. However, this Scrum rule (*no changes after sprint planning meeting*), while amplified by the two-month cycle, probably helped ConsultCo deal with the potential constant scope redefinition which was an important issue in the specific context of this project due to its previous unsuccessful efforts aimed at defining user needs.

Tailored Scrum and Multifaceted Agility in the Knowledge+ Project

When all the adaptations to the original Scrum method have been accounted for, is the resulting tailored system development approach still agile? Sarker and Sarker (2009) raise the issue; a "stripped" agile method may well end up looking like a traditional method. But they also argue that the ISD approach chosen is only one of the many potential ways a project team can be agile. It can be argued that the adjustments need to be done in a systemic way while taking into consideration the context. If an "agile practice" is impossible to apply in a certain context, careful consideration needs to be given to the practices that will be adopted to assess if overall the selected practices contribute or hinder the team's agility. This process is in and of itself an expression of agility.

So if agility is more than the ISD method, what if agility is an attitude as proposed by Cockburn (2002)? First, it is possible to observe that ConsultCo management team's desire to design an agile method able to deal with the specificity of the Knowledge+ project is already a sign of agility. Suffice it to say that it was a hard sell to top management and to the project management office and a person was passed over as project director because he could not see the value in managing a risky project in such a non-conventional way. Second, questioning and resisting the conventional way of managing projects and taking the less travelled road of designing a new approach based on the team's collective expertise and knowledge to fit a specific client's context is also a solid sign of agility. Overall, this case is a prime example of agility as an attitude.

The level of agility can further be explored through the three agility dimensions of Sarker and Sarker (2009). While the relative importance of each dimension and their subcomponents is still up for debate, a project team that possesses the three types of agility in sync with its context is most likely more "agile" than the one who has none of the practices that support the agility dimensions. In this line of thought, a careful analysis of the Knowledge+ project shows that the tailoring efforts, as presented in Table 2, cover all three dimensions of agility. Table 3 shows that tailoring efforts to deal with distributed work were the most frequent and are related to all three

agility dimensions. The majority of tailoring efforts related to distance (see Table 2, efforts 2 to 8) are associated with process-based agility and covers all three subcomponents (method, environmental awareness and temporal bridge-based). In addition, distance tailoring efforts are made to increase resource-based agility (effort 5) and linkage-based agility (effort 7) as well. Finally, working with a global client required Scrum tailoring in relation to resource-based agility, more specifically in relation to the people-based subcomponent.

Agility Dimensions	Description (from Sarker and Sarker, 2009)	Knowledge+ Project Key Tailoring Efforts (from Table 2)
Resource agility <i>*People-based</i> <i>*Technology-based</i>	Agility arising from the team's access to appropriate resources. -Agility arising from the access to 'multiskilled, flexible people' at the different locations for rapid deployment as and when necessary. -Agility enabled by the team member's seamless access to comparable, compatible, and relevant information and communication technology (ICT) infrastructure at each location.	<i>Working at a distance and with a global client</i> - A super user who had final authority on the group's decisions (9). - Technology was used as a substitute for face-to-face sprint reviews (5).
Process agility <i>*Method-based</i> <i>*Environmental-awareness based</i> <i>*Temporal bridge-based</i>	Agility arising from the team's project-related process-practices. -Agility arising from the appropriate adaptation of the traditional ISD methodology/practices, and proper harnessing of agile methods. -Agility arising from the systematic ability in the teams to anticipate, recognize, and react to changes in the project. -Agility arising from the ability of the team to collaborate across locations without significant temporal delays, and when possible, by taking advantage of time zones.	<i>Working at a distance</i> - The sprint cycle extended to two months (2). - The client insisted on having a set of fully documented functionalities before the development phase was launched (6). - For legal considerations and as per its ISO requirements, ConsultCo was eager to have documented and signed-off specifications that could be used as proof that the required functionalities had effectively been delivered, (8). - Every month, ConsultCo sent the team's two senior architects to meet face-to-face with the client's team on the west coast to discuss the content of the next sprint backlog (4). - Teams agreed on a fixed period of the day where each would be available for telephone and e-mail communications (3).
Linkage agility <i>*Cultural mutuality-based</i> <i>*Communicative relationship-based</i>	Agility arising from the nature of interactional relationships within distributed team and between the team and its customers. -Agility arising from the team members being part of a seamless mesh of cultures such that there is a minimal hindrance to the development of shared understanding and mutual relationships. -Agility arising from the ability to seamlessly communicate with each other, including continuous awareness of the actions/reactions of distributed team members, mature communicative relationships across locations, and seamless connectivity with customers.	<i>Working under a fixed-price contractual agreement umbrella and at a distance</i> - ConsultCo had a permanent representative based at the client's office on the west coast (7). - By gauging the level of complexity, ConsultCo made a time estimate for each functionality to be delivered by the end of the project. Each sprint initiation ended up being a mini-contract negotiation about the functionalities to be included, the level of sophistication awarded to each and how much effort each one would require (1).

Table 3: Overview of Findings – Tailoring Scrum and Agility Dimensions (Sarker and Sarker, 2009)

While Sarker and Sarker focus on distributed ISD, our findings show that their agility dimensions may help address other challenges experienced by ISD projects in various contexts. Firstly, in the absence of the detailed specifications of a conventional contract, it was said earlier that trust and relational governance were key elements to make the agile method work in a contractual relationship. Linkage-based agility highlights this need for flexibility and openness for an agile method to work. Secondly, Sarker and Sarker also include a technology-based ability to deal with distance. While the respondents in our case acknowledged the utmost importance of the communication tools they had at their disposal, they also considered essential their monthly trips to the client's site. In all the intricacies of building a long-term client-consultant relationship, they clearly preferred synchronous communication episodes over asynchronous ones. So as it was long ago discussed in the literature (e.g., Daft et al., 1987), we propose to add to the resource agility not only the issue of access to technology, but also the people's ability to choose the best communication tools for any given task. As discussed by Dubé and Robey (2008), in distributed work, the key to success may reside in the ability to carefully dose how face-to-face and each technology-based communication tool, that it be synchronous or asynchronous, are used. While access may be a necessary condition, it is not sufficient to ensure success.

Agility Dimensions	Knowledge+ Project Supporting Quotes ²
Resource agility <i>*People-based</i> <i>*Technology-based</i>	Working at a Distance and with a Global Client <i>"One very important point is that the client created a super user representative role empowered to make decisions. It's part of the client's culture because they have offices all around the world and the requirements and opinions can be very different. When we were face-to-face for our monthly meetings, the client was ready, able to answer our questions and to sign off the monthly requirements." (Functional architect and Scrum Master)</i> <i>"ConsultCo and the client deployed effective tools to share project documentation, programs, test results, and demonstrations. This technology was used as a substitute for face-to-face sprint reviews and accelerated communications." (Technical architect)</i>
Process agility <i>*Method-based</i> <i>*Environmental-awareness based</i> <i>*Temporal bridge-based</i>	Working at a Distance <i>"Within the fixed-contract agreement, it is important to work closely with the client and refine the priorities on a monthly basis. By the end of the project, the client will end up with the system he decided to have, not the system he asked for at the beginning." (Project director)</i> <i>"Every month, we negotiated with the client regarding the use cases estimates planned for the next sprint. It became easier as months went by because both the client and the consulting teams developed a shared understanding of cases that often ended up being quite similar." (Account manager)</i> <i>"The key players in the project [from the consulting side], the Scrum Master and the architects, work face-to-face with the client on a monthly basis. We adopted a consulting approach; we are not a coding factory. We built with the client a relationship based on trust. During the first two sprints, we delivered as promised and the client was impressed. He asked for Scrum training to develop a better understanding of how we were working." (Functional architect and Scrum Master)</i> <i>"It's critical to be on the client's site on a regular basis. You have to know the people and to see how they work... to understand and to anticipate the problems. It's impossible to work only at a distance in this type of project." (Functional architect and Scrum Master)</i> <i>"At the end of a sprint, 80% of my time was spent on working at aligning the expectations and managing the testing issues. Many e-mails and conference calls. Everybody had to be aligned and to support the technical decisions." (Technical architect)</i>
Linkage agility <i>*Cultural mutuality-based</i> <i>*Communicative relationship-based</i>	Working under a Fixed-Price Contractual Agreement Umbrella and at a Distance <i>"Working at a distance means more communications than in a traditional on site project. In our project, key team members had very good verbal communication skills using video, audio conference, and phone. With a Scrum approach, we work under short cycles and because written communications may often lead to misinterpretations, we prefer verbal contact." (Functional architect and Scrum Master)</i> <i>"We have a permanent on-site strategic consultant managing the relationship between our firm and the client. It helps to understand what's going on, to anticipate, to feel the perceptions. It was key for the project. In addition, I travel to the client's site every 4 to 6 weeks to talk to the people, to manage issues, and to anticipate situations and opportunities. It is also a way to gather information and to give a solid feedback to my local team." (Project director)</i> <i>"Working at a distance in this project means that we have many interfaces to manage with the client who has defined specific managerial, functional, and technical roles. Communication is critical for this project and I made this fact very clear with all my team members. For example, before sending an e-mail, try to use other verbal or face-to-face communication media. Write only to confirm, not to ask questions." (Project director.)</i>

Table 4: Overview of Findings – Supporting Quotes

² All interviews were conducted in a foreign language, so all quotes reported in this section were translated into English, omitting all information that could lead to the identification of the respondents and the organizations involved.

Overall, the tailoring efforts and quotes from the key players in the Knowledge+ project presented in Table 4 support Sarker and Sarker's (2009) findings that "agile methods need suitable adaptation for effective adoption/use in a distributed ISD setting... and that the application of agile methods is just one of the potential sources of achieving methodology-based agility" (p. 459). In the last part of their paper, Sarker and Sarker are looking at establishing the relative importance of each agility dimension/subcomponent. Extending their reflection, since most organizations do not respect all the rigid precepts of a single ISD method, maybe it is the overall performance of the designed method in dealing with the specificity of the context at hand this is of utmost importance. Maybe a method is agile if it fits perfectly the project context at hand, whatever the context may be. This seems to support the vision of agile as an attitude.

CONCLUSION AND CONTRIBUTIONS

Many papers have addressed the limitations of the traditional software development approaches and proposed new software development methods, including agile methods. However, these promising methods are based on a constraining set of assumptions. Their applicability to software development efforts involving different contexts and different types of application still poses important challenges. This study investigated the specific challenge of developing systems in the context of a global software development teams working at a distance and showed how a large consulting firm successfully tailored an agile method to develop a strategic financial application in a global development context involving teams from different organizations and countries in a contractual consultant-client relationship. The tailoring efforts addressed the challenges related to three distinct contextual areas. The challenges of *working under a fixed-price contractual agreement* were addressed by creating a total effort umbrella and by continuously aligning the functionalities to be delivered (as defined in the contract) with the work to be realized at the beginning of each sprint. Thus, each sprint represented a mini-contract negotiation process based on trust and shared knowledge between the client's and the consultant's teams. The challenges of *working at a distance* were addressed by adding time to the typical one-month sprint and by replacing daily face-to-face time with the client with a rich portfolio of communication tools and practices, such as daily technology-based communications, intensive monthly face-to-face meetings at the client's location, and the presence of a permanent consultant's representative based at the client's office overlooking the overall partnership between both parties. The challenges of *working with a global client* were addressed by introducing smart and useful communication-oriented documentation during a sprint (a "just enough" strategy), supporting the communication and approval processes between the client's team members and between the client's and the consultant's teams. Moreover, the efforts undertaken to adapt Scrum to the fixed-price contractual agreement context and to working with a global client are related to both the linkage and the resource agility dimensions, respectively. The tailoring efforts needed to work with a distant partner were related to all three agility dimensions.

Implications for Research

Anecdotal evidence and experience reports by agile method advocates are legion. Recently, IS researchers claimed that despite the growing popularity of agile approaches, we still do not know much about how agile methods can be adapted to address the specific needs of different project contexts and a plea was made to gather rigorous empirical evidence aiming at a better

understanding of their benefits, challenges, domains of applicability, and tailoring. This article contributes to the goal of rigorously building a cumulative knowledge base in the information system development field by providing empirical evidence as to how a leading world class consulting firm tailored Scrum to successfully develop a complex financial system for a global client located thousands of miles away. The findings show how the project team, inspired by Scrum, designed a method well suited to address the specific challenges of their organizational, contractual, and project contexts. This study answers Sarker and Sarker's (2009) call to study the nature of different work arrangement contexts and agility in distributed information systems development. Our study provides some empirical support for the usefulness of defining agility in a distributed IS development environment as a multifaceted concept, as defined by Sarker and Sarker. Our findings however show that the agility dimensions may be addressing challenges that go beyond distance and that agility as an attitude may be a powerful way to think about agility that go beyond investigating the rigid precepts prescribed by agile methods. More research efforts need to be put into studying how together ISD practices lead to agility, in the defined constraints of a specific context.

Implications for Practice

This study provides practitioners, and more specifically IS/IT consulting firms, with a roadmap of how to tap into the potential of agile methods and to tailor them to their specific needs. Agile methods emerged as a viable solution to overcome many limits of traditional software development approaches and are said to positively influence system development and to be particularly well suited for contexts where requirements change during the development life cycle. However, these methods, and Scrum in particular, faces new challenges related to increased specialization and globalization of system development activities; now, they bring together global client organizations working with global IT consulting firms through relationships formally defined by contractual agreements.

Consequently, distance tends to be increasingly part of the system development project contexts and make using agile methods more complex. The global client raises the issue of the identification and the legitimacy of the product owner and the challenge of defining the requirements so as to satisfy a large number of stakeholders in different geographic regions of the world.

The case of the Knowledge+ project provides guidelines for the development of good practices related to the tailoring of agile methods to address the specific needs of working at a distance with a global client while dealing with the constraints of a contractual agreement. The case of the Knowledge+ project shows how to overcome some limitations of agile methods by relaxing a constraining set of assumptions. Building a relationship based on open communication, knowledge sharing, and trust, and defining roles and responsibilities in a clearly defined process seem to be key elements. The findings also show that all the tailoring efforts in the Knowledge+ project are related to the agility dimensions, dimensions which together may provide practitioners involved in a contractual consultant-client relationship with a global client in a distributed environment with a useful framework to pinpoint the challenges and identify ways to deal with them. Agility as an attitude, as displayed by the local branch of ConsultCo, may also prove itself to be a powerful source of inspiration. Finally, the findings provide practical insights that can

serve as a model for consulting firms that evolve in complex environments and face increased competition. In this particular case, ConsultCo's members credit their innovative tailored approach as the source of their competitive edge and the reason they won this contract.

Limitations and Future Research Directions

Findings from a single case study and adopting an approach focused on the consultant's point of view cannot be generalized and the contextual factors unique to our case are likely to have influenced the manner in which the agile method was tailored. However, according to Orlikowski and Baroudi (1991), "the intent is to understand the deeper structure of a phenomenon, which it is believed can then be used to inform other settings" (p. 5). Future research should observe tailoring practices and their outcomes in multiple consulting organizations involved in development projects with global clients. The assessment of the tailoring actions' relative importance in different development contexts, the characteristics of the client-consultant relationship, the competencies required for successful agile tailoring, and the project management leadership styles are all relevant questions to examine in order to develop a better understanding of the evolution of the agile system development approaches in the context of accelerated globalization of both the clients and the IS/IT consulting firms.

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ABOUT THE AUTHORS

Carmen Bernier is Associate Dean, Academic and Student Affairs, and an Information Systems professor at the Faculty of Business Administration, Laval University. She holds a Ph.D. in Business Administration from HEC Montréal. Her current research focuses on information technology management in large organizations. Her work has been published in leading international journals, such as *Systèmes d'information et management*, *Data Base for Advances in Information Systems*, *Canadian Journal of Administrative Sciences*, *Gestion*, *Revue internationale de gestion*, and *Revue internationale de c@s en gestion*.

Line Dubé is a professor in the Department of Information Technologies at HEC Montréal. She holds a Ph.D. in Business Administration from Florida International University. Her current research focuses on different facets of IT project management and she has a clear preference for the richness of qualitative methods. Her work has been published in many journals, such as *Information Technology & People*, *MIS Quarterly*, *Communications of the ACM*, *Journal of Organizational Change Management*, *Information Systems Journal*, *Gestion*, and *Journal of Information Science*. She is also the author of many IT teaching cases published in the *International Journal of Case Studies in Management*, three of which have won the annual best case award. She is a member of the editorial board of *Information and Organization*.

Vital Roy is an honorary professor at HEC Montréal. He holds a Ph.D. in Business Administration from HEC Montréal. His work has been published in leading international journals such as *Systèmes d'information et management*, *Data Base for Advances in Information Systems*, *Canadian Journal of Administrative Sciences*, *Gestion*, *Revue internationale de gestion*, *Information Systems Journal*, and *International Journal of Case Studies in Management*.

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